

ANA SOFIA SILVA PINTO

INFORMATICS AND COMPUTING ENGINEERING

Tech explorer with a love for full-stack development, backend engineering, and the crossroads between cybersecurity and AI. I enjoy designing systems that are not only functional but resilient and future-proof.

As I continue my Master's in Network and Information Systems Engineering, I'm looking for new challenges where I can learn fast, contribute with impact, and push boundaries.

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EDUCATION

PRESENT STUDIES

MASTER'S IN NETWORK AND INFORMATION SYSTEMS ENGINEERING

Faculty of Sciences, University of Porto (FCUP) | 2024-2027 (Expected)

- Advanced networking, distributed systems, cybersecurity

PREVIOUS STUDIES

Bachelor in Informatics and Computing Engineering

Faculty of Engineering, University of Porto (FEUP&FCUP) | September 2021 - July 2025

- Relevant coursework: Algorithms, Software Engineering, Web Development, Databases, IA, Parallel and Distributed Computing, Computer Networking

WORK EXPERIENCE

SOFTWARE DEVELOPMENT INTERN

Visma Tech Portugal - Porto, Portugal

July 2025 - August 2025

- Contributed to the development of VisionPlanner, a SaaS product for accounting, by implementing backend features in .NET (C#) and integrating frontend components in React.
- Delivered two real user stories, including a critical bug fix for access control and a new feature for automatic user role assignment across distributed services.
- Participated in Scrum ceremonies (planning, dailies, reviews), wrote unit tests, and collaborated via Git, Azure DevOps, and Jira in a professional agile environment.

TECHNICAL SKILLS

- **Languages:** C/C++, Java, Python, C#, TypeScript, JavaScript, PHP, Dart, Haskell
- **Frameworks & Libraries:** .NET, React, Flutter
- **Backend & Databases:** Microservices, Distributed Systems, REST APIs, MySQL, Docker
- **Tools & Methodologies:** Git, Azure DevOps, Jira, CI/CD, Agile/Scrum
- **Areas of Interest:** Cybersecurity, Artificial Intelligence
- **Spoken Languages:** Portuguese (*Native*), English (*Fluent*), Spanish (*Basic*)

PROJECTS

SprintHive - a Project Management Platform (PHP, JavaScript, MySQL)

- Developed a **full-stack project** management platform with PHP (Laravel), JavaScript, and MySQL to streamline team collaboration and task management.
- Designed features for **project creation, task assignment, and team communication**, applying **agile methodologies** throughout the development lifecycle.

Java-- Compiler (Java, ANTLR, Jasmin, JVM)

- Built a **full, three-stage compiler from scratch** for a Java subset, managing the entire pipeline from front-end lexing/parsing (ANTLR) and intermediate representation (OLLIR) to back-end Jasmin bytecode generation.
- **Implemented rigorous semantic analysis** to guarantee code correctness and applied optimizations to generate efficient, executable bytecode for the JVM.

Distributed Chat System (Java, LLM)

- Architected a **high-concurrency, secure chat system** leveraging Java 21 virtual threads for optimal performance.
- **Secured communications** using SSL/TLS encryption (javax.net.ssl) and ensured robust, thread-safe operations with ReentrantLocks.
- **Integrated a local LLM** (via Ollama) to create AI-powered chat rooms capable of contextual, automated replies.

More projects can be seen on my [Github](#) page.

ADDITIONAL INFORMATION

- **Interests:** Cybersecurity challenges (CTFs), AI, inspired by music + pop culture.